LHCb Open Data Analysis of Three-Body B-Meson Decays

Invariant-mass reconstruction, signal classification, CP-asymmetry diagnostics and Dalitz-plot studies

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HEP portfolio project

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Physics Motivation

- The visible Universe is matter dominated.
- CP violation is required but not yet sufficient in the Standard Model.
- Heavy-flavour decays probe weak phases and hadronic interference.

Why B mesons?

- many decay channels
- displaced vertices
- rich multibody phase space

LHCb Experiment and Dataset

- LHCb: forward spectrometer at the LHC.
- Optimized for beauty and charm hadrons.
- Excellent tracking, vertexing and particle identification.

Input sample

- CERN Open Data record 4900 (LHCb, 2017, CC0)
- ROOT files, `DecayTree` TTree
- track momenta, charge, PID, vertex variables
- 5.1×10^6 candidates

Analysis Workflow

ROOT → `uproot` → `pandas` → reconstruction → cuts → ML → physics observables

Data

Branch loading
candidate tables
reproducible paths

Physics

invariant mass
event selection
charge split

Outputs

figures
ML diagnostics
CP and Dalitz studies

Invariant-Mass Reconstruction

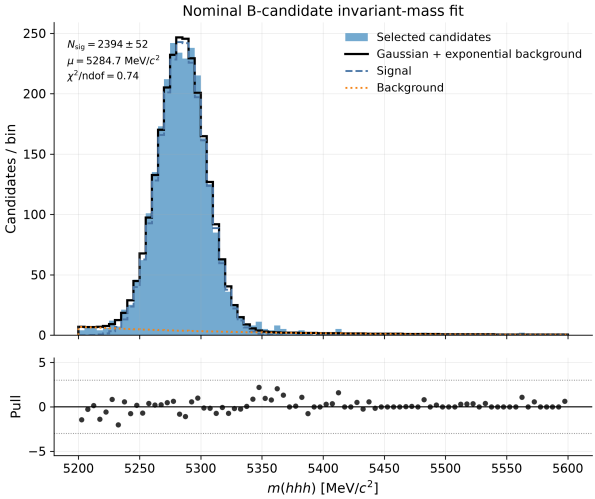
Three-kaon hypothesis

$$E_i = \sqrt{p_{x,i}^2 + p_{y,i}^2 + p_{z,i}^2 + m_K^2}$$

$$m_B^2 = \left(\sum_i E_i \right)^2 - \left| \sum_i \vec{p}_i \right|^2$$

- assign $m_K = 493.677\ \mathrm{MeV}/c^2$
- build daughter four-vectors
- sum to reconstruct B candidate
- vectorized in `numpy`/`pandas`

Mass Spectrum and Signal Fit



$$f(m) = N_s\, G(m;\mu,\sigma) + N_b\, E(m;\alpha)$$

- extended binned likelihood, MINUIT
- both polarities, 2573 candidates
- $N_s = 2394 \pm 52$
- $\mu = 5284.7 \pm 0.4$ MeV/c^2
- $\sigma = 19.5 \pm 0.4$ MeV/c^2

Fitted from 5200 MeV/c^2 up: below that lies a partially reconstructed structure the background model does not describe.

Event Selection

Baseline cuts

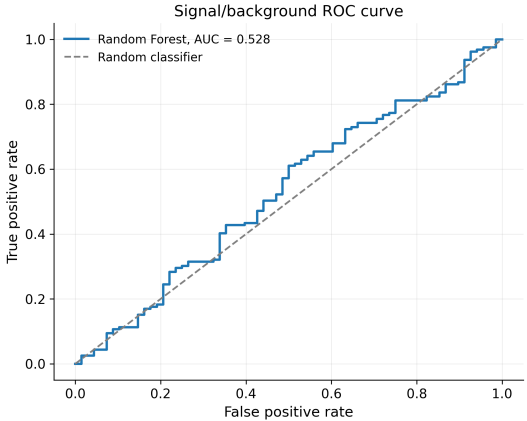
$$\chi^2_{\text{vtx}}(B) < 5$$

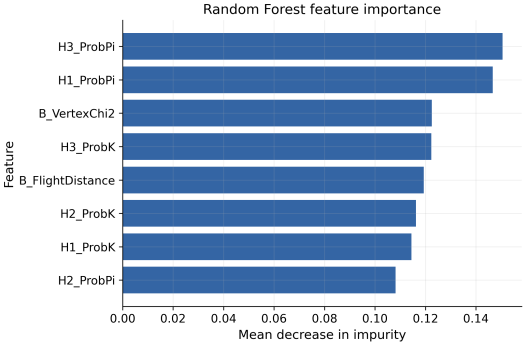
$$d_{\text{flight}}(B) > 20$$

$$P_K(H_1), P_K(H_2), P_K(H_3) > 0.5$$

- vertex quality: common displaced decay point
- flight distance: B lifetime signature
- kaon PID: consistency with $B \to KKK$
- same cuts reused across notebooks

Signal/Background Classification





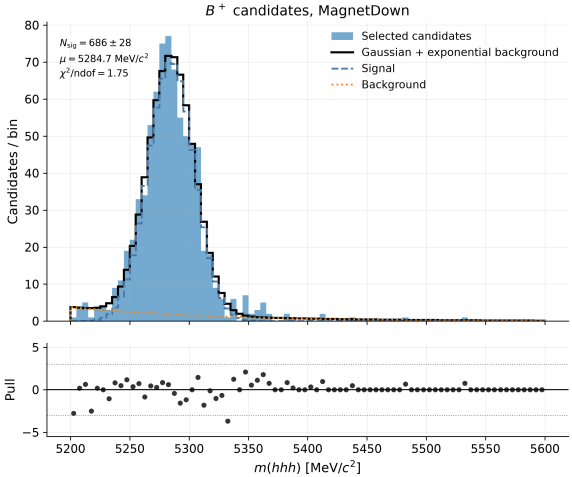
- Sideband-labelled Random Forest; mass excluded from the features.
- $\mathrm{AUC} = 0.528$: **no separation**. A negative result, reported as one.

The baseline cuts already removed the background the available features could discriminate, and only 755 labelled candidates remain. The classifier is not used in any result that follows.

Raw Charge Asymmetry

$$A_{\text{raw}} = \frac{N(B^-) - N(B^+)}{N(B^-) + N(B^+)}$$

- B^+ and B^- fitted *simultaneously*: shared shapes, own yields
- A_{raw} is a fit parameter
- Down -0.034 ± 0.029, Up -0.031 ± 0.032
- equal-weight polarity average cancels left–right detection asymmetries

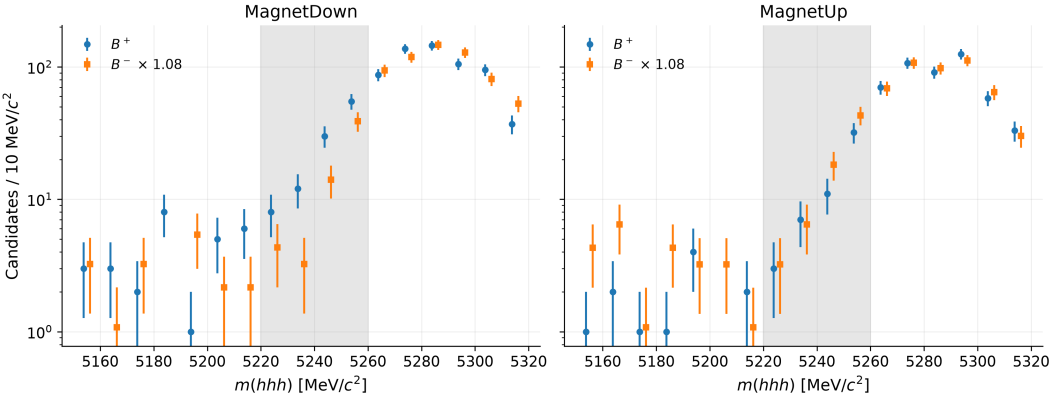


Result

$$A_{\text{raw}} = -0.032 \pm 0.021\,(\text{stat.}) \pm 0.016\,(\text{syst.})$$

Raw: no production, kaon-detection, trigger or efficiency corrections.

A Shoulder in One Spectrum: Real or Fluctuation?



- MagnetDown B^+: 105 candidates in 5220–5260 for 91 expected
- shape test, toy-calibrated: $p = 0.043$
- independent MagnetUp: no excess, $p = 0.25$
- a $\sim 2\sigma$ fluctuation, which separate fits had absorbed into the B^+ background

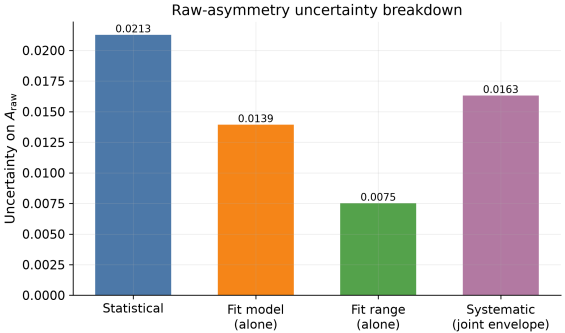
Validation and Systematics

2000 toy experiments

- both polarities generated, refitted, averaged
- pull width 0.98, bias 2.6% of σ
- no fit failures
- goodness of fit: Down $p = 0.025$, Up $p = 0.30$

Systematics

- 4 models $\times$ 5 ranges, both polarities
- envelope, not quadrature: 0.016
- range shifts all within $1.5\,\sigma_{\mathrm{Barlow}}$

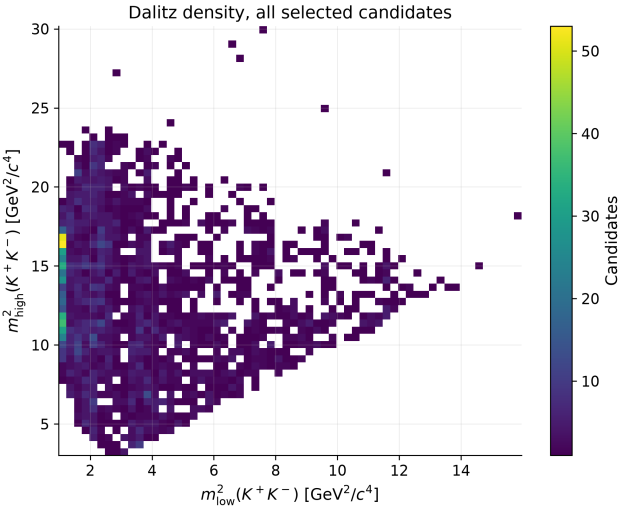


Configurations accepted only if every fit reaches a valid minimum with $\chi^2/\mathrm{ndof} < 2.5$ per charge spectrum — criteria fixed in advance. All 20 pass; MagnetUp barely moves across the grid.

Dalitz-Plot Analysis

- Pair masses
  $m_{ij}^2 = (E_i + E_j)^2 - |\vec{p}_i + \vec{p}_j|^2$
- Track labels carry no meaning: pairs found from the charges
- Opposite-sign pairs, folded:
  $$m^2_{\text{low}}(K^+K^-) \leq m^2_{\text{high}}(K^+K^-)$$

Same coordinates for B^+ and B^- under CP; the $\phi(1020)$ band sits at $m^2_{\text{low}} \approx 1.04\ \text{GeV}^2$.



Limitations and Future Work

Current limitations

- raw asymmetry: production and kaon detection uncorrected
- momentum scale not calibrated per charge
- low-mass structure excluded, not modelled
- classifier does not separate

Next steps

- control channel for production/detection asymmetries
- per-charge momentum-scale calibration
- partially reconstructed shape from simulation
- background subtraction per Dalitz bin

Conclusion

- Reproducible HEP workflow from ROOT ntuples to physics plots.
- Invariant-mass reconstruction under a kaon mass hypothesis.
- Shared selections and modular Python functions.
- Likelihood fits, toy validation, systematics and Dalitz diagnostics.
- $A_{\mathrm{raw}} = -0.032 \pm 0.021\,(\mathrm{stat.}) \pm 0.016\,(\mathrm{syst.})$, raw and uncorrected.

A compact portfolio analysis mirroring a real LHCb workflow